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Breathing patterns and CO₂ production in adult spiny mice (*Acomys cahirinus*)

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Abstract

The spiny mouse (*Acomys*) is a precocial mammal with unique regenerative abilities. We used whole-body plethysmography to describe the breathing patterns and CO₂ production (VCO₂) of adult spiny mice (n = 10 male, 10 female) and C57BL/6 mice (n = 9 male, 11 female). During quiet breathing, female but not male spiny mice had lower tidal volumes and CO₂ production vs. C57BL/6 mice. During extended hypoxia (30 min), male and female spiny mice decreased VCO₂ and tidal volume to a greater degree than C57BL/6 mice. During an acute hypoxic-hypercapnic respiratory challenge (10% O₂, 7% CO₂), male and female spiny mice had blunted ventilatory responses as compared to C57BL/6 mice, primarily from a diminished increase in respiratory rate. These data establish a baseline for studies of respiratory physiology and neurobiology in spiny mice in the context of their remarkable regenerative capacity and their unique background of a desert dwelling species.

Keywords

Comparative physiology; Respiratory physiology; Regenerative medicine; Spiny mice

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Conflict of interest

The authors declare no competing financial interests.

CRediT authorship contribution statement

SR, CSS, DDF: Conception or design of the work; SR: Acquisition of data; SR, MDS, JG, CSS DDF: Analysis or interpretation of data for the work; SR, DDF: Drafting the work or revising it critically for important intellectual content; SR, MDS, JG, CSS DDF: Final approval of the version to be published; SR, MDS, JG, CSS DDF: Agreement to be accountable for all aspects of the work.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.resp.2022.103975.

1. Introduction

The spiny mouse, *Acomys cahirinus*, is a precocial mammal that displays unique regenerative abilities in adulthood. Notably, spiny mice demonstrate complete regeneration of dermis and cartilage after ear lesions due to a “pro-regenerative environment” that maintains cellular proliferation after the lesion (Seifert et al., 2012). More recent studies indicate that after spinal cord injury in the spiny mouse, spinal inflammation and fibrosis are attenuated (Streeter et al., 2020) and neural regeneration is enhanced (Nogueira-Rodrigues et al., 2021). Thus, *Acomys cahirinus* and other *Acomys* species are important organisms for comparative physiological studies that may provide insights regarding the physiologic mechanisms required for mammalian regeneration (Gaire et al., 2021). In addition, owing to their precocial development, spiny mice have been studied as a model for diabetes, renal physiology, reproductive physiology, and perinatal development (Maden and Varholick, 2020).

The purpose of the present study was to use whole-body plethysmography to measure breathing patterns in male and female spiny mice and to use respirometry to measure carbon dioxide production (VCO_2). The spiny mouse is an important comparative physiology model (Gaire et al., 2021; Maden et al., 2021), and here we compared and contrasted spiny mouse breathing to values measured in a standard laboratory mouse strain (C57BL/6). Breathing was measured during “eupnea” as well as acute respiratory challenges with hypoxia (10% O_2) or combined hypoxia-hypercapnia (10% O_2 ; 7% CO_2). Since differences in breathing are usually found when rodent strains are directly compared (Han and Strohl, 2000; Hodges et al., 2011; Strohl et al., 1997) we hypothesized that breathing patterns and responses to respiratory challenge would differ between spiny vs. C57BL/6 mice. Since to our knowledge breathing has not previously been carefully evaluated in the adult spiny mouse, this study is intended to establish a baseline for studies of respiratory physiology neurobiology in spiny mice in the context of their remarkable regenerative capacity, and provide foundational datasets upon which to study the impact of neuromuscular injuries on breathing in this unique comparative mammalian model.

2. Materials and methods

2.1. Experimental animals

All experiments were conducted in 4-month-old spiny mice ($n = 20$, 10 males ($44.9 \text{ g} \pm 2.9 \text{ g}$) and 10 females ($37.6 \text{ g} \pm 3.4 \text{ g}$); *Acomys cahirinus*; in-house colony at the University of Florida) and C57BL/6 mice ($n = 20$, 9 males ($31.3 \text{ g} \pm 3.4 \text{ g}$) and 11 females ($24.5 \text{ g} \pm 4.3 \text{ g}$); Mus; Jackson Laboratory). All animal work conducted in this study is in accordance with the guidelines provided by the United States Department of Agriculture (USDA) and was approved by Institutional Animal care and use committee (IACUC) at the University of Florida. Animals were communally housed in a controlled environment (12 hr light/dark cycles) with *ad libitum* access to food and water.

2.2. Whole body plethysmography

Unrestrained and unanesthetized animals were studied in a flow-through whole-body plethysmograph chamber (approximately 400 ml in size) to measure breathing (Rana et al., 2021). Animals were maintained at an ambient temperature in the range of 23–24 °C, which falls within the suggested housing temperature range for spiny (21–26 °C) and C57BL/6 mice (25.5–27.6 °C) (Haughton et al., 2016; Keijer et al., 2019). A small proportion of expired air outflow (<50cc/ minute) from the plethysmograph chamber was split through a CO₂ sensor (Cat # T6615–10KF, DigiKey; measurement range: 0–10,000 ppm) to quantify metabolic activity based on Fick's principle (Fick, 1870). Freely moving and awake animals were acclimated to the plexiglass recording chamber two days in a row for an hour each day. Plethysmography recordings were conducted between 9 am and 5 pm. Air flow was maintained at 2 liters/minute through the plethysmography chamber. On the third day, during a 40-minute baseline period, mice (n = 12 spiny (6 M, 6 F); n = 12 C57BL/6 (6 M, 6 F), were exposed to normoxic air (21% O₂, 79% N₂) and recordings were made using a Buxco Finepointe system sampled at 500 samples per second (500 Hz). Animals were subsequently challenged to a 5 min hypoxic (10.5% O₂, 89.5% N₂) period, followed by a 10-minute recovery period under normoxic air, ending with a 5 min hypoxic-hypercapnic respiratory challenge (maximal chemoreceptor challenge; 10.5% O₂, 7% CO₂ balanced N₂). In a subset of animals (n = 8 spiny (4 M, 4 F); n = 8 C57BL/6 (3 M, 5 F), prolonged effects of hypoxia exposure were studied by exposing animals to a 30-minute hypoxia (10.5% O₂, 79% N₂) gas challenge. Ventilation was measured by monitoring pressure swings within the plexiglass chamber caused by gas expansion and contraction due to the heating and humidification of air as the animal breathes during this resting state. Tidal volume was quantified using the Drorbaugh and Fenn equations (Drorbaugh and Fenn, 1955). VCO₂ was measured using Fick equation $VCO_2 = (FECO_2 - FICO_2) * V/[1 - (FECO_2 * (1 - (1/RQ)))]$ with an assumed RQ of 0.85 based on assuming a mixed source of fats, carbohydrates and proteins as being used as metabolic substrate and animals displaying a moderate amount of baseline activity.

Body temperature was collected at the end of the recording protocol by placing mice in a small cone and using a rectal temperature probe. Repeated measurements of body temperature in spiny mice are challenging due to the fragile nature of their skin that tears easily upon restraint. In a small subset of animals (n = 3 F spiny and n = 3 C57BL/6 mice), we measured changes in body temperature from the quiet baseline period (30–35 min after placing animal in chamber) to the end of the hypoxia-hypercapnia period and from the start and end of the extended hypoxia protocol (n = 3 F C57BL/6; n = 3 F spiny). These additional experiments were done to determine how much body temperature varied across the plethysmography recording time.

2.3. Data analysis

Data for plethysmography traces were averaged over 5 min of baseline breathing. During the 5-minute hypoxia and hypoxic-hypercapnic respiratory challenge, data was averaged during the last 3 min of the challenge, when the diaphragm muscle is expected to elicit maximal response for that respective behavior. Furthermore, data from the 30-minute hypoxia trial was averaged for consecutive 5-minute windows for the duration of the trial. Data for breathing challenges is represented as normalized to the baseline normoxic activity for

that particular animal in order to reduce inter-animal variability and emphasize the peak ventilatory response as a percent change from baseline.

Plethysmography data collected using spike2 software was exported to MATLAB (The MathWorks, Natick, MA), where analysis of waveforms was conducted using a custom software. Plethysmography tidal volume traces were filtered using a Butterworth filter (0.5–6 Hz) to reduce EKG artifact. Primary respiratory outcomes included tidal volume (ml/kg), respiratory rate (bpm, breaths per minute; determined by counting the number of breaths detected within the designated analysis window period and dividing by the duration in minutes), minute ventilation (V_E , ml/min/kg, product of the respiratory rate and tidal volume), CO_2 production (VCO_2 ; instantaneous VCO_2 values averaged over analysis window period) and V_E/VCO_2 (ratio of metabolic activity and minute ventilation).

2.4. Statistical Analysis

All statistical evaluations were performed using JMP statistical software (version 15.0, SAS Institute Inc., Cary, NC). Power calculations for each cohort of animals in the study were determined prior to initiation of study and based on pilot experiments. The study statistical design was powered to consider the expected standard deviation in tidal volume measures (20%) and in order to detect a 25% difference at a power of 0.9 and alpha of 0.05. Using the mixed linear model, statistical significance was established at the 0.05 level and adjusted for any violation of the assumption of sphericity in repeated measures using the Greenhouse-Geisser correction. Group (spiny mice, C57BL/6), sex (male, female) and timepoints (baseline and hypoxia (0–5, 5–10, 10–15, 15–20, 20–25 and 25–30 min) were included as model variables, using animals as a random effect. When appropriate, *post hoc* analyses were conducted using Tukey-Kramer honestly significant difference (HSD). Normality of the distribution was assessed using the Shapiro-Wilk test within each species. Data in text and figures are presented as mean and standard error (SE), unless otherwise specified.

3. Results

Representative respiratory traces recorded using whole-body plethysmography C57BL/6 and spiny mice are shown in Fig. 1. Note the increase in peak inspiratory airflow during hypoxia and the hypoxia-hypercapnia or maximal chemoreceptor challenge (“Maxchemo”). A summary of all statistical tests is provided in Table 1. Body temperature data are summarized in Tables 2 and 3. There was no difference in body temperatures across spiny and C57BL/6 mice ($p = 0.96$). Female C57BL/6 mice had a lower body temperature as compared to their male counterparts ($p = 0.024$). Spiny mice exhibited a 0.1 °C increase in body temperature from the baseline to the end of the hypoxia/hypercapnia exposure, while C57BL/6 mice exhibited a 0.3 °C decrease over the same period. Relative to baseline, body temperature decreased after the 30-minute hypoxia protocol by 0.8 °C for spiny mice and 0.7 °C for C57BL/6 mice.

3.1. Baseline “eupneic” measurements

Mice were first studied during quiet breathing (Fig. 2.). CO₂ production at baseline showed an interaction between group and sex ($p = 0.024$). Post hoc analysis indicated that female spiny mice had lower CO₂ production than female C57BL/6 mice ($p = 0.02$). When comparing within group, there was no difference in CO₂ production between female and male spiny mice ($p = 0.50$), but female C57BL/6 mice had a higher CO₂ production as compared to male C57BL/6 mice ($p = 0.013$). Minute ventilation (V_E) also showed a group x sex interaction ($p = 0.01$), with spiny females having lower V_E as compared to C57BL/6 females (0.047). The ratio of V_E to CO₂ production (V_E/VCO_2) was reduced in females ($p = 0.036$ vs. male), but there was no group effect ($p = 0.426$). Inspection of the data, however suggests a possible group x sex interaction (Fig. 2C, $p = 0.055$). Respiratory rate was different between males and females ($p = 0.049$) with no effect of group ($p = 0.387$). Baseline tidal volume showed group x sex interaction (Fig. 2E, $p < 0.001$) with female spiny mice having lower tidal volume compared to C57BL/6 females ($p < 0.001$). Female spiny mice showed a trend for a reduced tidal volume compared to males ($p = 0.058$). In contrast, female C57BL/6 mice had greater tidal volume when compared to males ($p < 0.001$).

3.2. Response to acute hypoxia

The peak hypoxic ventilatory response was assessed over the last 3 min of a 5-minute exposure (Fig. 3). The most striking observation was that the increase in minute ventilation was considerably lower in spiny mice as compared to C57BL/6 mice. Thus, there was a significant effect of group ($p < 0.001$) but no effect of sex ($p = 0.1604$) or group x sex interaction ($p = 0.95$). The reduced hypoxic ventilatory response was mirrored in the V_E/VCO_2 data with an effect of group ($p = 0.004$) and sex ($p = 0.04$) but no group x sex interaction ($p = 0.62$). V_E/VCO_2 also tended to be higher in female spiny and C57BL/6 mice as compared to males.

3.3. Response to sustained hypoxia

We also assessed the ventilatory response to a more prolonged (30 min) hypoxia exposure (Fig. 4 and Supplemental Table 1). Upon exposure to hypoxia, all mice displayed a brief (~ 1 min) period of arousal where they became more active (Fig. 4A). This coincided with a small increase in CO₂ production in both C57BL/6 and spiny mice. However, both groups then dropped their CO₂ production considerably by 6–10 min. The decrease was more pronounced in spiny mice, especially after 20–30 min. Statistical analyses of CO₂ production during prolonged hypoxia revealed a significant group x time interaction ($p = 0.0015$).

Minute ventilation showed a significant group effect during prolonged hypoxia ($p = 0.045$), indicating differences between C57BL/6 and spiny mice. All mice showed an initial increase in V_E upon hypoxia exposure. Values returned to baseline levels by minute 6–10 in C57BL/6 mice, and remained at baseline for the duration of the recording. In contrast, V_E dropped to 50–60% of baseline values in spiny mice.

There was no significant effect of group on V_E/VCO_2 during sustained hypoxia ($p = 0.086$). However, the following trends were noted. C57BL/6 mice showed a larger increase in $V_E/$

VCO₂ (~55% baseline) in response to hypoxia which plateaued after 5–10 min. Spiny mice displayed a lower increase in V_E/VCO₂ (~30% baseline) which peaked at 10 min, followed by a small decline. V_E/VCO₂ also varied significantly with time during the 30-minutes exposure ($p < 0.001$). Post-hoc analysis revealed that while the increase in V_E/VCO₂ from baseline was significant at the 0–20 min points, the decline during 21–30 min was not statistically different from baseline or the 0–20 min timepoints.

Respiratory rate was similar between spiny and C57BL/6 mice during the prolonged hypoxia exposure with no effect of group ($p = 0.77$). There was, however, an effect of time ($p < 0.001$) on respiratory rate. An acute increase was observed in both groups followed by a gradual return to baseline levels after approximately 10 min.

Tidal volume during hypoxia showed a group x time interaction ($p < 0.001$). C57BL/6 mice showed an initial increase in tidal volume during hypoxia. This response was blunted in spiny mice, which did not increase tidal volume in the initial 5-minute window. In C57BL/6 mice, the initial increase in tidal volume returned to baseline levels by 5 min. However, in spiny mice, tidal volume was further depressed past baseline and dropped to nearly 50% of baseline by the 30-minutes time point.

The frequency of augmented breaths (sighs) was evaluated during the hypoxia exposure, and a group x time interaction was observed (Fig. 5, $p < 0.001$). C57BL/6 mice displayed a substantial increase (~400% of baseline) in sigh frequency in the initial 5-minute hypoxic period that decreased and plateaued at ~200% of baseline sigh incidence by 6–10 min. Spiny mice also displayed an acute increase in sigh incidence (albeit lower as compared to C57BL/6 mice). This increase returned to baseline sigh incidence by the 6–10-minute timeframe.

3.4. Maximal ventilatory challenge

The ventilatory response to a “maximum chemoreceptor challenge” was achieved using exposure to a hypoxia-hypercapnia gas mixture (Fig. 6). During this challenge, spiny mice increased V_E as expected, but the increase was less than what was observed in C57BL/6 mice (group effect, $p < 0.001$). In addition, male mice had lower ventilatory responses as compared to females (effect of sex; $p = 0.026$), but there was no group x sex interaction ($p = 0.488$). The blunted ventilatory response in spiny mice occurred partly due to reduced breathing frequency compared to C57BL/6 mice (group: $p < 0.001$). Breathing frequency tended to be lower in male mice (sex effect, $p = 0.067$) with no indication of a group x sex interaction ($p = 0.627$). The reduced ventilation during the max challenge in spiny mice was also associated with a blunted tidal volume as compared to C57BL/6 mice (group: $p = 0.002$). The tidal volume response was lower in males (sex: $p = 0.028$) with no group x sex interaction ($p = 0.172$).

4. Discussion

The spiny mouse is a unique model for comparative physiology, primarily due to its remarkable regenerative capacity (Gaire et al., 2021). The present study provides the first comprehensive evaluation of breathing and CO₂ production in the adult spiny mouse. The

spiny mouse showed a few different respiratory and/or metabolic outcomes as compared to the widely used C57BL/6 laboratory mouse. During quiet breathing (i.e., “eupnea”), female spiny mice utilized a shallow breathing pattern. During an extended hypoxic exposure (30 min), male and female spiny mice had reduced CO₂ production and tidal volume as compared C57BL/6 mice. The spiny mice also showed a reduced response to the very strong respiratory stimulus provided by a dual hypoxic-hypercapnic challenge. Overall, these results establish a baseline for future studies of this highly unique mammal, and suggest possible differences in respiratory control mechanisms between the spiny mouse and C57BL/6 mouse.

4.1. Comparative respiratory physiology and breathing in the spiny mouse

Differences in breathing are to be expected when different rodent strains are directly compared (Han and Strohl, 2000; Hodges et al., 2011; Strohl et al., 1997). By studying these differences, the genetic underpinnings of respiratory control mechanisms can be elucidated (Gaultier, 2004). The C57BL/6 mouse is a widely studied model, and has been included in several prior comparative respiratory studies (Gonsenhauser et al., 2004; Han and Strohl, 2000). For example, C57BL/6 J mice have a rapid-shallow breathing pattern as compared to C3H/HeJ mice. C57BL/6 J mice also show a greater hypoxic and hypercapnic ventilatory response (Tankersley et al., 1997). Segregation analyses of the inheritance pattern from this study also suggested that phenotypic differences in breathing frequency and inspiratory time could be mapped to as few as two genetic loci. Other studies have reported that C57BL/6 mice breathe faster, shallower and with lesser variability as compared to A/J mice, despite exhibiting minimal differences in mechanics of the respiratory system (Han and Strohl, 2000; Han et al., 2001).

Here we compared breathing and CO₂ production between spiny and C57BL/6 mice under standardized laboratory conditions. To our knowledge, the only prior respiratory physiology study in the spiny mouse was conducted using an in vitro experimental preparation. That study of neonates was designed to determine if the precocial newborn spiny mouse would be a more mature model of respiratory rhythm generation as compared to a standard laboratory mouse model (Greer et al., 1996). The results showed that P0-P1 spiny mouse brainstem-spinal cord preparations did not produce a stable respiratory rhythm, but the isolated medulla slice produced a respiratory rhythm quite similar to what was observed in Sprague-Dawley rats. In the current study, we did not observe any striking differences in baseline eupneic breathing patterns between spiny and C57BL/6 mice. However, some sex-specific differences were noted. Specifically, female spiny mice generated lower relative tidal volumes compared to female C57BL/6. Female spiny mice also displayed lower CO₂ production as compared to female C57BL/6 mice at baseline, suggesting a lower basal metabolism. In this regard, it is interesting to note that spiny mice have recently been identified as the first and only menstruating rodent species (Bellofiore et al., 2018). Although cyclical changes and hormonal regulation in this unique mammalian reproductive system is still an under-studied area, we speculate that observed differences in the control of breathing between female spiny and C57BL/6 mice could originate from differences in sex hormones. Indeed, common ovarian hormones such as progesterone are known

respiratory stimulants (Behan et al., 2003). These sex-specific differences further emphasize the uniqueness of the spiny mouse model and its utility as a comparative physiology model.

4.2. Respiratory phenotype during gas challenges

Physiological responses to acute and chronic hypoxia have been widely studied in many rodent models (Teppema and Dahan, 2010). Burrowing rodents are known to be more tolerant to hypoxic conditions, and this is achieved by matching metabolic demand with O₂ delivery (Li et al., 2021). For example, hypoxic exposure can lead to a reduction of metabolic rate, known as hypoxic hypo-metabolism, thereby helping to match metabolic demand to O₂ availability. On the other hand, hypoxia will also produce an increase of ventilation, particularly during acute phases, thereby increasing O₂ delivery. Tankersley et al. showed that C57BL/6 mice respond to acute hypoxia (3–5 min, 10% O₂) by increasing ventilation, but with relatively little change in their metabolic rate (Tankersley et al., 1994). In contrast, we found that while C57BL/6 increased CO₂ production during acute hypoxia. This increase in metabolic activity was particularly pronounced in female mice, which was not included in a prior study (Tankersley et al., 1994). However, during prolonged exposure to hypoxia, the metabolic activity dropped to ~80% of baseline levels by the 6th minute of the hypoxic exposure and remained suppressed for the duration of the exposure. This was matched with a return of tidal volume to baseline levels by the 6th minute. Thus, C57BL/6 solve the problem of acute hypoxia by increasing tidal volume, and then during longer exposures, respond primarily with a decrease in CO₂ production. Spiny mice also decreased CO₂ production during extended hypoxia, and did so to a much greater extent than the C57BL/6 mice. This feature is also common amongst the naked mole rat, a fossorial burrowing mammalian species which drastically reduces metabolic rate in response to acute and chronic hypoxic exposures (Farhat et al., 2020; Ilacqua et al., 2017). Further, spiny mice showed a considerable decrease in tidal volume as the time in hypoxia progressed. Thus, spiny mice may provide a new and unique comparative model of hypoxic ventilatory adaptations during acute and prolonged exposures.

A very strong chemoreceptor stimulus was provided in the current studies by exposing the mice to a combined hypoxic-hypercapnic gas mixture (Tankersley et al., 1994). During these conditions, the spiny mice showed smaller increase in both respiratory rate and tidal volume as compared to C57BL/6 mice. This observation may indicate that spiny mice have altered chemosensory mechanisms that allow them to adapt to their unique desert-dwelling lifestyle. In particular, there have been reports of spiny mice living in underground burrows of other rodents and termite mounds, areas which contain relatively low O₂ and high CO₂ air compositions (Deacon, 2012).

Another phenotypical breathing observation was the reduced number of augmented breaths (“sighs”) in spiny mice during the extended hypoxic period as compared to C57BL/6 mice. Augmented breaths occur periodically during eupneic breathing, are normal airway protective behaviors that may serve to prevent lung atelectasis and maintain effective lung function (Housley et al., 1970). The blunted sigh frequency response in the spiny mouse may represent another unique respiratory control adaptation.

4.3. Technical considerations

The impact of core body temperature on calculations of respiratory tidal volume during whole body plethysmography has been discussed in detail (Mortola and Frappell, 1998). In the current study, core body temperature was only collected once at the end of the recording period due to technical challenges presented with the use of spiny mice (Gaire et al., 2021). In particular, the skin of spiny mice can tear easily upon restraint, and tail handling is not advised (Haughton et al., 2016). In our study we used a cone to temporarily restrain spiny mice to take temperature without directly scruffing their skin or tail. However, since our estimations of tidal volumes were made based on a temperature measure taken at the end of the recording protocol, this is a potential source of error. In a subset of animals, we observed that spiny and C57bl/6 mice exhibited a 0.1–0.3 °C change in core body temperature from baseline period to the end of the hypoxia/hypercapnia period. As expected, we also observed a slightly larger decrease in body temperature at the end of the 30-minute hypoxia protocol (0.8 °C in spiny mice and 0.7 °C in C57BL/6 mice). Based on the Drorbaugh and Fenn equation used to calculate tidal volume (Drorbaugh and Fenn, 1955), a change in body temperature of 1°C, with all other variables held constant, would result in an ~6.5% change in the estimate of tidal volume. Future studies can benefit from implantable telemetric body temperature measuring probes for a real time assessment of changes in body temperature, and thus a more accurate calculation of respiratory volumes via whole body plethysmography.

In the current study, rate of VCO_2 production was used as a representative of metabolic rate. However, the most accurate values would require the measurement of both VO_2 and VCO_2 concentrations through inspired and expired air flow and the use of equation from Frappell et al. that accounts for changes in total flow associated with O_2 consumption and CO_2 production (Frappell et al., 1992). Since our testing system was not equipped to collect O_2 concentrations, we calculated VCO_2 with an assumed RQ of 0.85 using the Fick's equation. Furthermore, this equation has been shown to produce an error of $\pm 3\%$ if the actual RQ is 0.7–1.0 (Withers, 1977).

4.4. Conclusion

The current study is the first to characterize breathing and CO_2 production in the adult spiny mouse which is a unique mammal with remarkable regenerative properties (Gaire et al., 2021). The results provide insight into the adaptive responses of this emerging laboratory rodent model, in comparison to a commonly used laboratory mouse strain, C57BL/6 mice. Spiny mice appear to have a blunted respiratory response to ventilatory challenges that perhaps arises from the physiological adaptations in their body to the hostile desert conditions. Overall, these results provide a baseline for studies of respiratory physiology and neurobiology in spiny mice as well as studies of neurologic injury (Nogueira-Rodrigues et al., 2021; Streeter et al., 2020).

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data Availability

Data will be made available on request.

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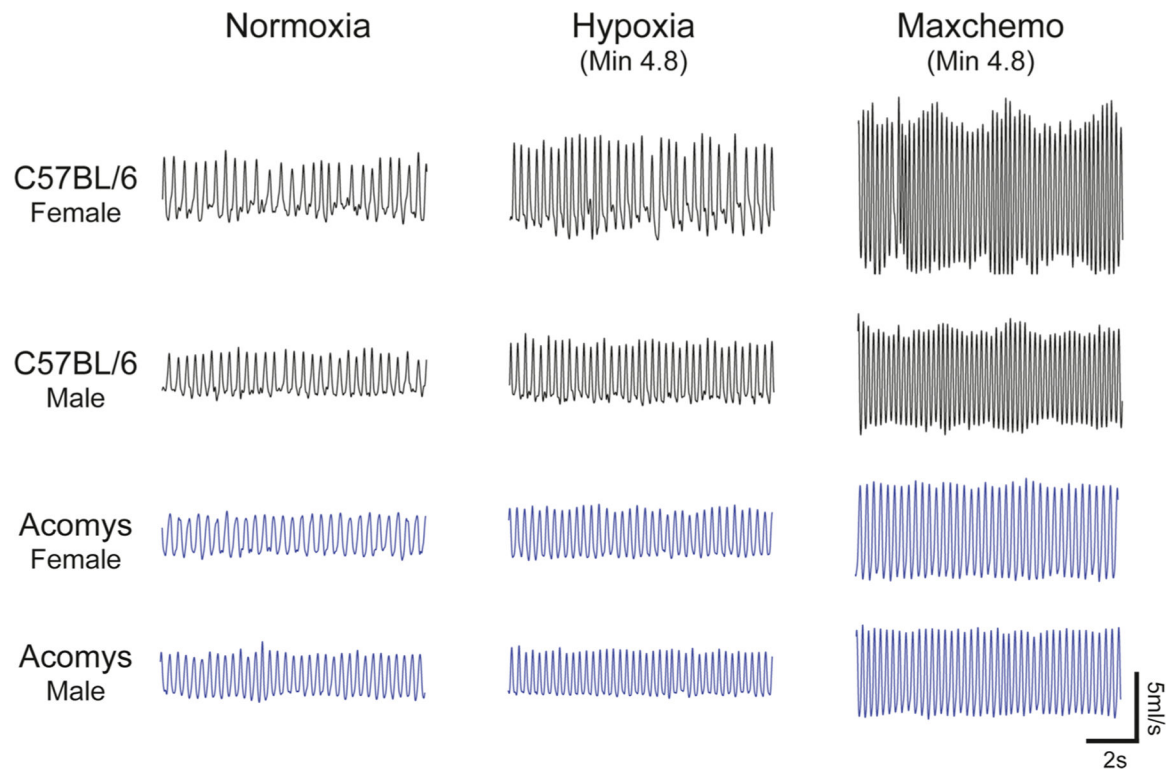
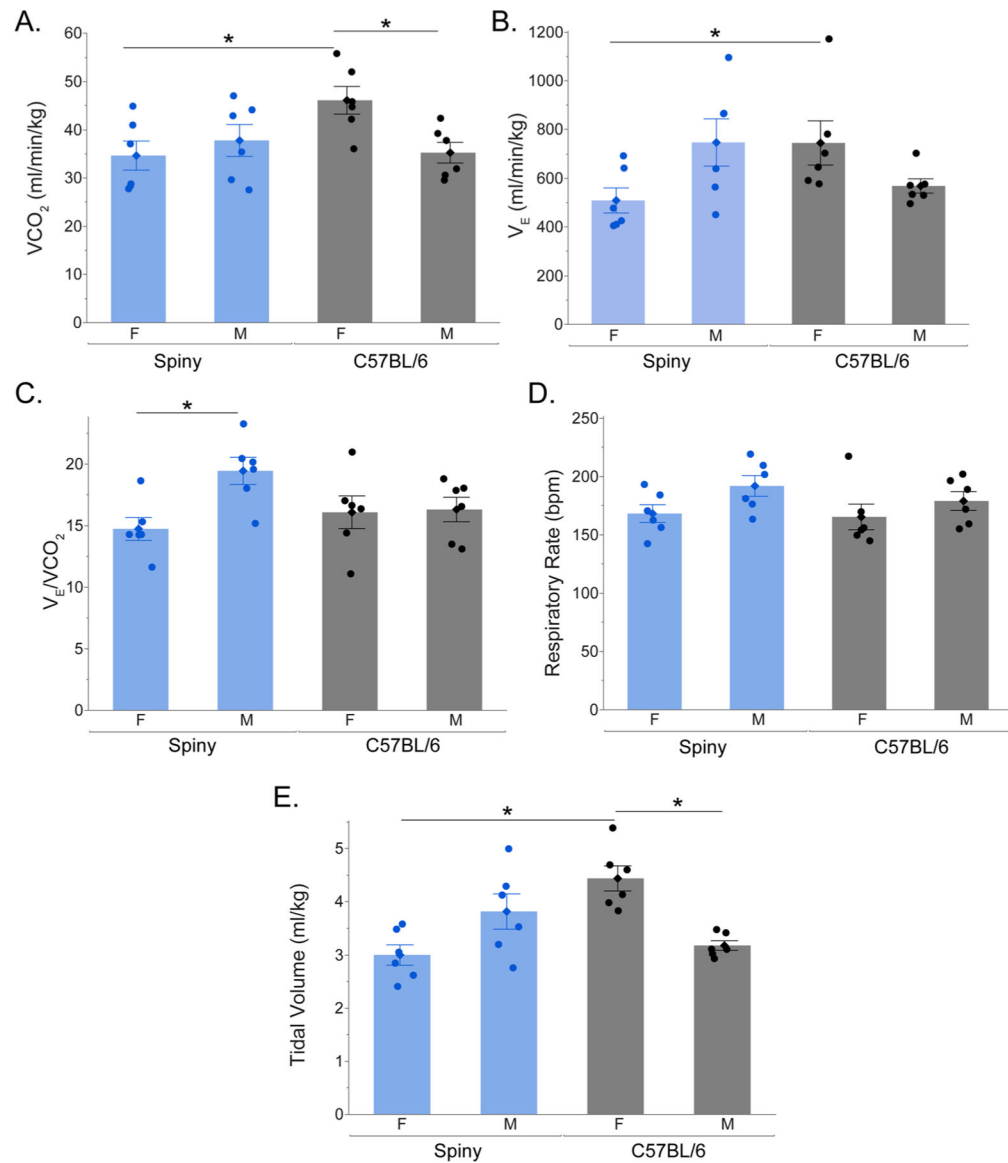
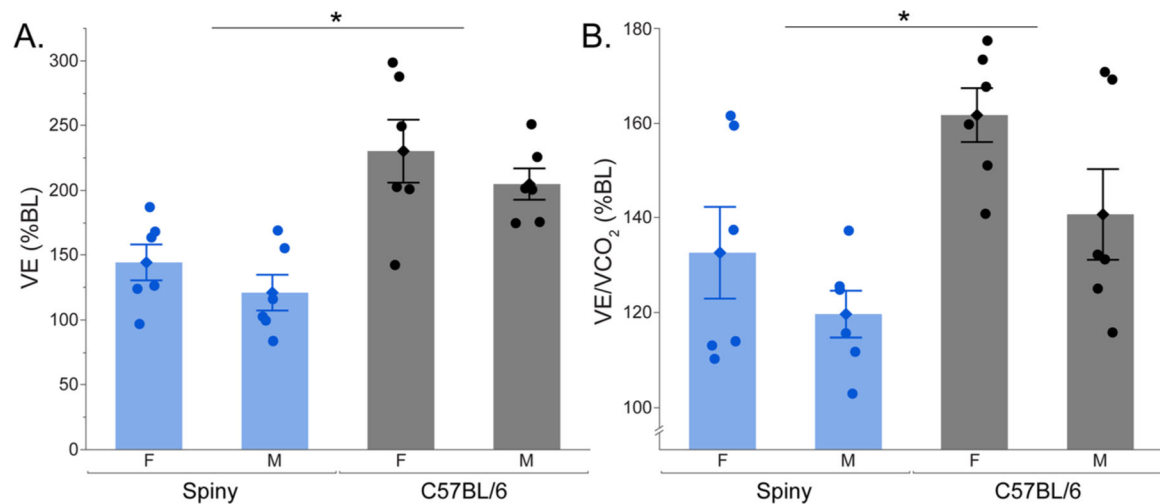


Fig. 1.

Breathing in adult C57BL/6 and Acomys. Representative examples of air flow recorded using whole-body plethysmography in unanesthetized C57BL/6 (black) and Spiny (blue) mice. Ten-second-long respiratory flow traces are shown from animals during quiet baseline breathing of normoxic air (21% O₂, 79% N₂), minute 4.8 of hypoxic air (10.5% O₂, 79% N₂) and minute 4.8 of maximal chemoreceptor challenge (10.5% O₂, 7% CO₂ balanced N₂). Inspiration is indicated by an upward deflection in the trace. Scaling on each trace is the same between the C57BL/6 and spiny panels.

**Fig. 2.**

Mean ventilatory parameters during eupneic breathing. Respiratory parameters from a 5-minute baseline period are summarized. Female spiny mice had significantly lower V_{CO_2} and tidal volumes compared to female C57BL/6 mice. Female spiny mice also had a lower ratio of minute ventilation to CO_2 production (V_E/V_{CO_2}) ratio compared to male spiny mice. (A) V_{CO_2} (ml/min/kg); (B) V_E (ml/min/kg); (C) V_E/V_{CO_2} ; (D) respiratory rate (breaths per minute; bpm); (E) tidal volume (ml/kg); Statistical interactions for this data are summarized in Table 1. * $p < 0.05$. Data are presented as bar plots with all individual data points corresponding to individual animal means. C57BL/6 mice are depicted in black and spiny mice are depicted in blue. Group means are shown in diamond with error bars depicting $\pm$ SE.

**Fig. 3.**

Mean ventilatory parameters during peak hypoxic response. Respiratory parameters from minute 3–5 during a hypoxic exposure (10.5% O₂, 79% N₂) are summarized. Spiny mice show a blunted response to a hypoxic challenge in both minute ventilation (V_E) and V_E/VCO_2 (expressed as a % of baseline). Female spiny and C57BL/6 mice tended to have a higher V_E and V_E/VCO_2 compared to their male counterparts during the challenge. A) V_E (expressed as % of baseline); (B) V_E/VCO_2 (expressed as % of baseline); Statistical interactions for this data are summarized in Table 1. * $p < 0.05$. Data are presented as bar plots with all individual data points corresponding to individual animal means. C57BL/6 mice are depicted in black and spiny mice are depicted in blue. Group means are shown in diamond with error bars depicting $\pm$ SE.

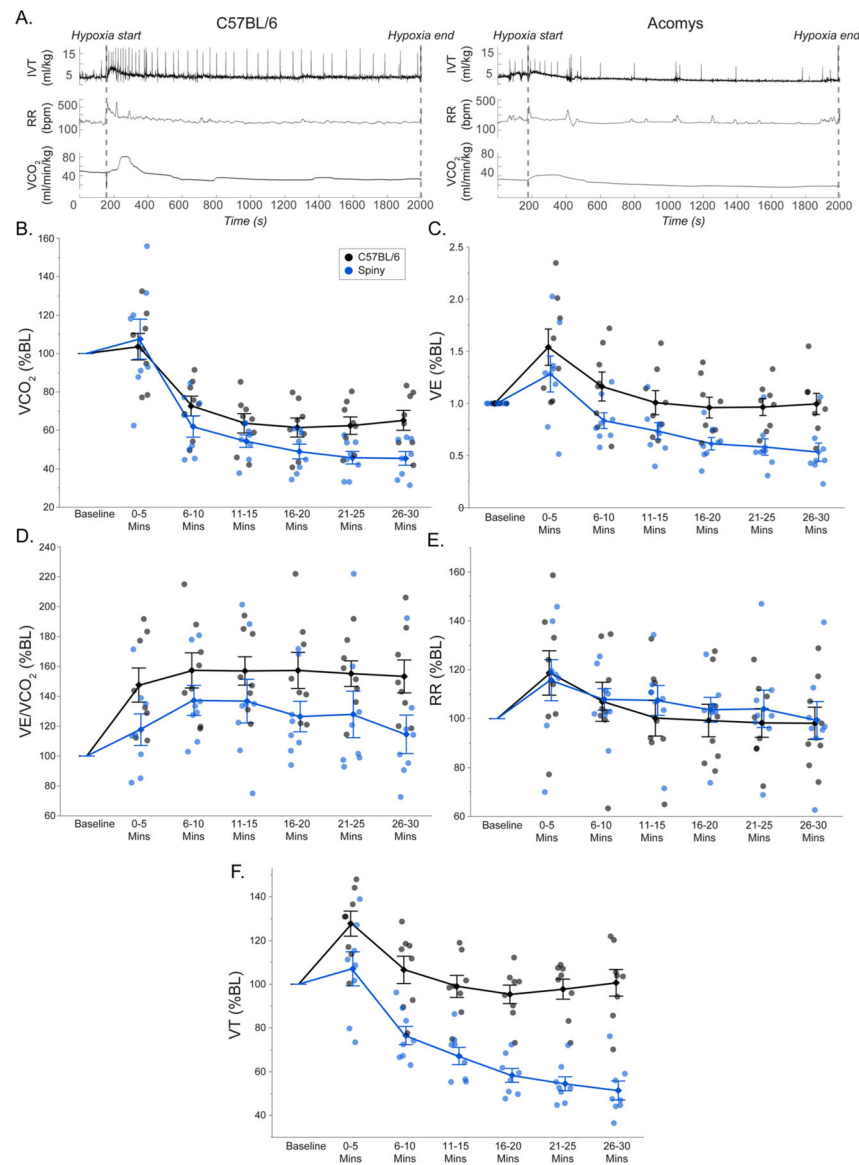


Fig. 4. Mean ventilatory parameters during extended hypoxic response (30 min). A. Representative examples of instantaneous tidal volume (IVT), instantaneous respiratory rate (RR) and VCO₂ from C57BL/6 (left) and spiny (right) mice are shown from an extended hypoxic exposure period, preceded by a short baseline period. Note the sharp increase in IVT, RR and VCO₂ during the initial phase of the exposure (more pronounced in C57BL/6 mice). Large spikes in IVT depict incidences of spontaneous sighs (deep breaths). B.–F. Mean ventilator data during the extended hypoxic exposure is depicted. Both C57BL/6 and spiny mice drop their CO₂ production during their hypoxic exposure, albeit more pronounced in spiny mice. Spiny mice show a blunted response to the hypoxic exposure (a drop in tidal volume and a smaller increase in V_E/VCO₂). Statistical interactions for this data are summarized in Table 1. Data are presented as individual data points corresponding to

individual animal means. C57BL/6 mice are depicted in black and spiny mice are depicted in blue. Group means are shown in diamond with error bars depicting $\pm$ SE.

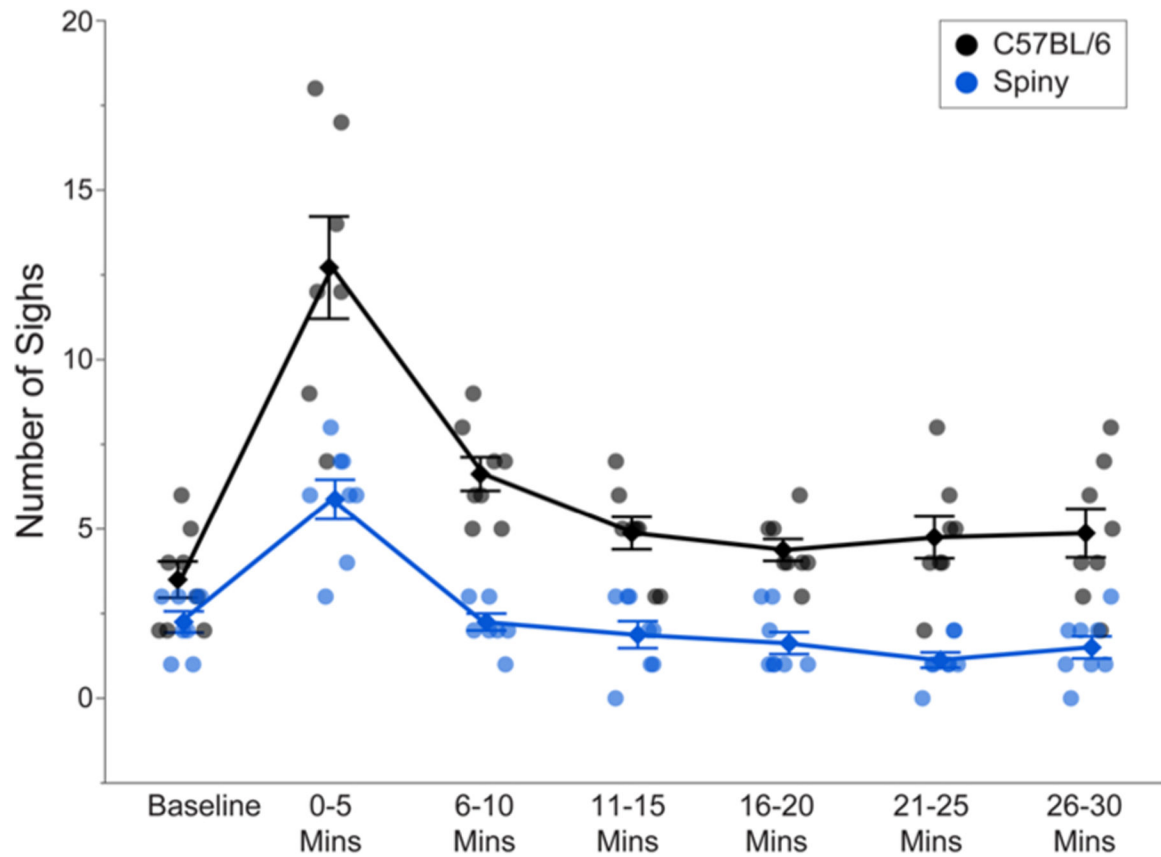


Fig. 5. Incidence of deep breaths during extended hypoxia. An increase in frequency of spontaneous deep breaths (sighs) was observed during the initial 5-minute window of the hypoxia exposure. In the extended hypoxic phase, the incidence of sighs in spiny mice dropped back to baseline levels but remained elevated in C57BL/6 mice. Statistical interactions for this data are summarized in Table 1. Data are presented as individual data points corresponding to individual animal means. C57BL/6 mice are depicted in black and spiny mice are depicted in blue. Group means are shown in diamond with error bars depicting $\pm$ SE.

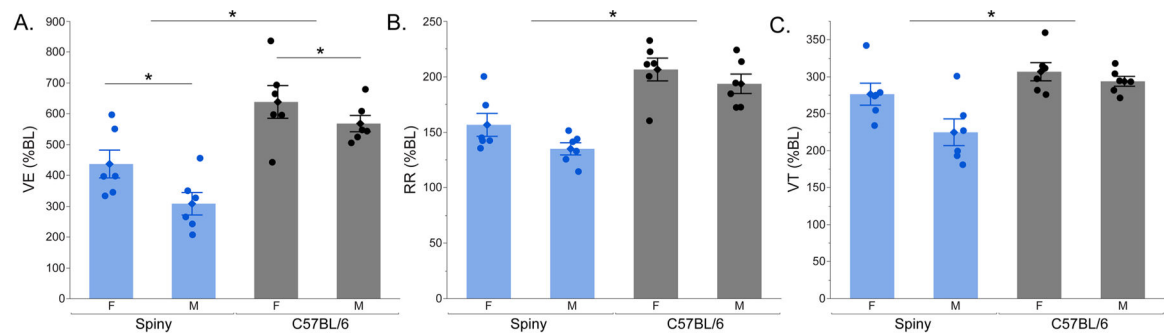


Fig. 6.

Mean ventilatory parameters during maximal chemoreceptor challenge. Respiratory parameters from minute 3–5 during a maximal chemoreceptor challenge (10.5% O₂, 7% CO₂ balanced N₂) are summarized. Spiny mice show a blunted response to the maximal chemoreceptor challenge in all three parameters (expressed as a % of baseline): minute ventilation (V_E), respiratory rate (RR) and tidal volume (VT). Similar to the hypoxic challenge, female spiny and C57BL/6 mice tended to have a higher V_E , RR and VT compared to their male counterparts during the challenge. A) V_E (expressed as % of baseline); (B) RR (expressed as % of baseline); (C) VT (expressed as % of baseline); Statistical interactions for this data are summarized in Table 1. * $p < 0.05$. Data are presented as bar plots with all individual data points corresponding to individual animal means. C57BL/6 mice are depicted in black and spiny mice are depicted in blue. Group means are shown in diamond with error bars depicting $\pm$ SE.

Summary of all statistical tests. Fixed effects include: Group (Spiny, C57BL/6), sex (male, female) and timepoints (baseline and hypoxia (0–5, 5–10, 10–15, 15–20, 20–25 and 25–30 min). DF (between-groups degrees of freedom) and DFDen (within-groups degrees of freedom). Summary data presented in Figs. 2–6.

Figure	Outcome	Effect	DF	DFDen	F Ratio	P value
2. Baseline “eupneic” measurements	VCO ₂ (ml/min/kg)	Sex	1	20	1.8137	0.1931
		Group	1	20	2.4284	0.1348
	VE (ml/min/kg)	Sex*Group	1	20	5.9606	0.0241 *
		Sex	1	20	0.177	0.6785
		Group	1	20	0.1564	0.6967
		Sex*Group	1	20	8.123	0.0099 *
	Respiratory Rate (bpm)	Sex	1	20	4.3544	0.0499 *
		Group	1	20	0.7819	0.3871
	Tidal Volume (ml/kg)	Sex*Group	1	20	0.3142	0.5813
		Sex	1	20	0.9356	0.345
3. Hypoxic ventilatory response	VE/VCO ₂	Group	1	20	3.0522	0.096
		Sex*Group	1	20	20.6028	0.0002 *
		Sex	1	20	5.0424	0.0362 *
		Group	1	20	0.6604	0.426
	VE (%BL)	Sex*Group	1	20	4.1554	0.0549
		Sex	1	20	2.1254	0.1604
	VE/VCO ₂ (%BL)	Group	1	20	25.7898	<0.0001 *
		Sex*Group	1	20	0.0041	0.9497
		Sex	1	20	4.7622	0.0412 *
		Group	1	20	10.3786	0.0043 *
4. Prolonged (30 mins) hypoxia exposure	VCO ₂ (ml/min/kg)	Sex*Group	1	20	0.2645	0.6127
		Group	1	14	2.4131	0.1426
		Time	6	84	107.585	<0.0001 *
		Group*Time	6	84	3.9691	0.0015 *
	VE/VCO ₂	Group	1	14	3.4008	0.0864
		Time	6	84	13.7843	<0.0001 *

Figure	Outcome	Effect	DF	DFDen	F Ratio	P value
	VE (ml/min/kg)	Group*Time	6	84	1.9543	0.0814
		Group	1	11.97	4.9915	0.0453 *
		Time	1	11.97	0.0247	0.8777
		Group*Time	1	11.97	0.6592	0.4327
	Respiratory Rate (bpm)	Group	1	14	0.0863	0.7732
		Time	6	84	7.6964	< 0.0001 *
		Group*Time	6	84	0.6051	0.7255
		Group	1	14	29.579	< 0.0001 *
	Tidal Volume (ml/kg)	Time	6	84	65.3646	< 0.0001 *
		Group*Time	6	84	17.9024	< 0.0001 *
		Group	1	14	30.1723	< 0.0001 *
		Time	6	84	32.3605	< 0.0001 *
5. Sigh Incidence during Hypoxia	Number of Sighs (% BL)	Group*Time	6	84	6.4841	< 0.0001 *
		Sex	1	20	5.7463	0.0264 *
		Group	1	20	30.9776	< 0.0001 *
		Sex*Group	1	20	0.5	0.4877
6. Maximal ventilatory challenge	Minute Ventilation (%BL)	Sex	1	20	3.7531	0.067
		Group	1	20	36.9193	< 0.0001 *
		Sex*Group	1	20	0.2437	0.6269
		Sex	1	20	5.6013	0.0281 *
	Tidal Volume (%BL)	Group	1	20	13.239	0.0016 *
		Sex*Group	1	20	2.0013	0.1725

Table 2

Summary of core body temperature data from C57BL/6 and spiny mice during the full plethysmography protocol (normoxic baseline, hypoxia and hypoxia-hypercapnia; n = 12 C57BL/6 (6 M, 6 F); n = 12 spiny (6 M, 6 F)) and during the extended hypoxia protocol (n = 8 C57BL/6 (3 M, 5 F); n = 8 spiny (4 M, 4 F)). Data are presented as Mean $\pm$ SD.

Species	Protocol	Sex	Body Temp (°C)
C57BL/6	Full	F	36.7 $\pm$ 0.3
		M	37.2 $\pm$ 0.4
	Hypoxia	F	36.2 $\pm$ 0.2
		M	36.5 $\pm$ 0.4
Spiny	Full	F	36.9 $\pm$ 0.4
		M	37 $\pm$ 0.3
	Hypoxia	F	36.5 $\pm$ 0.3
		M	36.1 $\pm$ 0.2

Table 3

Summary of change in core body temperature of C57BL/6 and spiny mice from the baseline period of the full plethysmography protocol to the end of the hypoxia-hypercapnia period (n = 3 F C57BL/6; n = 3 F spiny) and from the start and end of the extended hypoxia protocol (n = 3 F C57BL/6; n = 3 F spiny). Data are presented as Mean $\pm$ SD. Negative numbers indicate a drop in temperature as compared to baseline or starting body temperature of protocol.

Species	Protocol	Change in Body Temp ($^{\circ}$ C)
C57BL/6	Full	-0.3 ± 0.3
	Hypoxia	-0.7 ± 0.4
Spiny	Full	0.1 ± 0.4
	Hypoxia	-0.8 ± 0.06